Multi-Agent Systems

Agenda

- Single-Agent vs Multi-Agent
- Multi-Agent Architectures

Let’s begin the discussion by answering a few questions.

Multi-Agent Quiz

What is a primary limitation of a single agent system?

A. It cannot generate text longer than 500 words

B. It struggles with task decomposition and parallel reasoning

C. It cannot access external APIs

D. It only works for conversational tasks

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Single-Agent vs Multi-Agent

Dimension	Single-Agent System	Multi-Agent System (MAS)
Architecture	One agent handles planning, execution, and validation	Multiple specialized agents collaborate with defined roles
Capability	Best for simple, linear tasks	Best for complex, multi-step, parallel workflows
Scalability & Reliability	Limited scalability, single point of failure	Scalable, modular, and more robust if well-designed
Coordination Requirement	No internal coordination needed	Requires communication protocols and task orchestration
Design Complexity	Easier and faster to implement	Requires structured design and architectural planning

Multi-Agent Quiz

A central AI agent receives a task, divides it into smaller tasks, assigns them to different agents, and then combines all their outputs before giving the final result. Which type of architecture is this?

A. Blackboard Pattern

B. Market-Based Architecture

C. Orchestrator-Worker

D. Graph-Based Architecture

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Multi-Agent Architectures

Architecture	Core Structure	Control Style	Best For	Key Limitation
Orchestrator Worker	One central controller + multiple workers	Centralized	Structured workflows, task pipelines	Central controller is a single point of failure
Hierarchical	Multi-level layered agents	Layered control	Large, complex systems with abstraction levels	Slower decision flow
Blackboard Pattern	Shared central knowledge space	Decentralized via shared memory	Collaborative problem solving	Can become messy without structure
Graph-Based	Agents connected as nodes & edges	Semi-distributed	Flexible, adaptive systems	Design complexity

Power Ahead!